

ARC Final Report - Discovery-Project - (DP)

In the interests of promoting awareness of the benefits of publicly-funded research, the ARC may draw on the material you provide about your research for reporting and promotional purposes.

Part A. PROJECT IDENTIFICATION

A1 Program DP

A2 Project ID DP0771252

A3 Administering organisation Macquarie University

A4 First year funded 2007

A5 Title of funded project
Cohomology enhanced: an application of enriched and higher categories

A6 1st named Participant (note: if these details are incorrect amend your GAMS record)

<i>Title</i>	<i>Em/Prof</i>
<i>First Name</i>	<i>Ross</i>
<i>Family Name</i>	<i>Street</i>
<i>Gams ID</i>	<i>J00649</i>
<i>Organisation</i>	<i>Macquarie University</i>
<i>Participant role</i>	<i>CI</i>
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<i>Work phone number</i>	<i>61 2 9850 8921</i>
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<i>Fax number</i>	<i>61 2 9850 8114</i>

A7 Other Participants

Title	FirstName	Family Name	GAMS ID	Organisation	Participant role
Prof	Michael	Johnson	X63139	Macquarie University	CI
Dr	Stephen	Lack	AB2183	University of Western Sydney	CI
A/Prof	Dominic	Verity	B26033	Macquarie University	CI

A8 For this Project, total ARC Funding requested 462827

A9 For this Project, total ARC Funding received 287199

A10 Number of years for which Funding was received 3

Part B PROJECT DESCRIPTION AND OBJECTIVES

B1 100 word Project summary (from the original project application)

Cohomology has been one of the most powerful tools in the mathematics of the twentieth century, finding applications in all areas of modern mathematics. It is a technique for understanding and classifying complex mathematical structures in simpler terms. The current project involves a radical expansion in scope of the information extracted from these mathematical structures, using the most recent advances in enriched and higher-dimensional category theory.

B2 Summary of original objectives of project

1. Further develop the theory of Batanin's globular operadic weak n -categories and the Street-Verity theory of simplicial weak omega-categories (weak complicial sets) to the point where they support enhanced cohomology theories in the sense outlined in Street's work on descent.
2. Develop an analogous approach to the classical theory of abelian cohomology, in the context of DG-category theory (categories enriched in chain complexes).
3. Demonstrate that each of these cohomologies classifies local structure and supports a coherent theory of stacks and torsors.
4. Provide a framework in which to formalize our notion of enhanced cohomology.
5. Provide comparison mappings between these contexts to demonstrate that the resulting theories are consistent wherever they are commonly applicable; in particular, to develop suitable simplicial nerve constructions from globular weak n -categories and to demonstrate that they map these structures to weak complicial sets (or more precisely, into its subcategory of simplicial weak omega-categories).
6. Formulate and establish a coherence theorem for simplicial weak omega-categories, demonstrating their relationship to suitable Gray-enriched structures.
7. Apply these investigations to the recasting of classification results in traditional abelian cohomology theory and to outstanding problems in mathematical physics, algebraic topology, categorical Galois theory, group cohomology and computer science.

Part C PROJECT OVER DURATION OF FUNDING

- C1 Were there significant changes to the Project or the objectives? ☐ No
If yes, please briefly describe the changes and explain why they were made.

For example, changes may have been made because the ARC funding was less than the requested funding.

- C2 Did anything affect the satisfactory and timely progress or completion of the Project?
☐ Yes
If yes, please briefly describe what happened.

1. We had planned for the involvement and advice of Professor G.M. Kelly. However, he died on Australia Day 2007.
2. CI Johnson was appointed Vice-President of Macquarie University's Academic Senate. This delayed his visit to Cambridge University and has slowed the work on the manuscript with van Glabbeeck, and Winskel.
3. A significant part of the project was carried into 2010 because of late appointments of research associates.

- C3 Fellows on Team Projects - Briefly describe the Fellow's contribution to the Project
(ARC Fellows, who are Participants in a team of investigators but not the first-named Participant (Chief Investigator of Fellow), should contribute to this question)

Part D PROJECT OUTCOMES AND IMPACTS

D1 Did the project meet its initial objectives or approved revised objectives? ☒ Yes

D2 Describe briefly the significance, results and outcomes of the project.

The main significance of our work on this project is that it has opened up new ground and new techniques to supply the steeply increasing demand for our branch of mathematics. Our field of expertise is enriched and higher category theory and our project was to direct those areas to an enhanced cohomology theory. Our results are being applied to the traditional target area of algebraic topology but also to algebraic geometry.

The most easily measured outcome of the project is the huge number of research publications over the four years: 2 authored books, 5 edited books, 2 book chapters, and 54 refereed scholarly papers (see Part G).

We will discuss our results in the context of each objective listed in Part B2.

1) Verity published two papers in this period developing the theory of weak complicial sets [G3C1#1 and #20] and a monograph [G3A1#2] on their strict cousins - these were cited as in preparation in the original application. He has also developed the "quasi-category" theory of these structures specifically to support enhanced cohomology along the lines described in this passage. This includes:

*) Development of model category structures to support a theory of complete Segal spaces in weak complicial sets as a context for developing the internal category theory needed to translate the work which has now been published in Street and Verity. While this is still in preparation, Verity has spoken extensively about this work at the ICTM in Portugal, at the Hyland-Johnstone Fest in Cambridge and in our Australian Category Seminar.

*) The work Verity spoke about in 2010 on Conduché functors of quasi-categories - which again addresses the foundations required by a quasi-categorical translation of Street and Verity [G3C1#37]. This leads to a notion of representably powerful for the complete Segal space context discussed in the last point.

*) Verity's most recent work on Higher Derivateurs logically includes work he has done developing a more 2-categorical approach to the theory of discrete fibrations of quasi-categorical structures than one might find in the acclaimed work of Lurie, for example.

We actually moved away somewhat from tackling the globular approach to these matters. This was caused in part by the non-arrival of a suitable research fellow during the first 2 years. However, this work has recommenced, under our DP10 Grant and Batanin's Discovery Grant, now that Richard Garner and Mark Weber are on board.

2) The importance of DG-categories is always lurking in the homotopy theoretical aspect particularly. However there is nothing explicit to report on this objective over the period.

3) Work on this is well advanced. The Street and Verity work [G3C1#37] on torsors and local structure classification has appeared. We found that we had to revisit that work before we could advance it to the weak complicial/quasi-categorical context. Much of the success in Objective 1 leads to adaptation of our torsor theory to the more complex weak complicial sets. While we have done a great deal on this, it still needs to be prepared for publication.

4) In fact this is well addressed by what we have said about objectives 1 and 3.

5) A start was made in [G3C1.17] by describing a nerve construction for bicategories which allowed a precise comparison with the simplicially defined weak 2-categories of Tamsamani and Simpson. Continuation of this work is belatedly underway under the goals of our DP10 Grant. It was delayed by the unavailability of an appropriate research fellow: see D3 for the latest news on this.

6) Verity has developed a coherence theorem which uses some of the infrastructure from objective 1 to show that every "equivalence saturated" weak complicial set is equivalent to the nerve of some category strictly enriched in equivalence saturated weak complicial sets. The constructions used in this work appear to arise again, in one form or another, in his work with Garner under objective 5. A paper on this topic is in preparation and was presented at the International Category Theory Meetings in Portugal 2008 and Cape Town 2009. As part of that work Verity developed a weak higher category of cobordisms; connections with apparently related results by Lurie at MIT must be further investigated.

7) We have developed a very wide range of applications and techniques which can broadly be regarded as applications of enhanced cohomology. These include publications such as [G3C1##2-5, 7, 10, 14, 22].

D3 Did the project lead to exciting new research directions, innovations and/or collaborations, and/or lay the foundations for new research and/or new partnerships? ☒ Yes

If yes, please describe briefly how.

We have achieved many of these specific objectives - but emergent opportunities have also allowed us to achieve well beyond the bounds of the original project.

Garner and Verity are working towards a method for building models for certain contractible globular operads from weak complicial sets - and this work seems to be quite promising, although it is still at a very early stage of development.

D4 Are there identifiable national benefits—including economic, social, cultural and/or environmental contributions—resulting from this project? ☒ Yes

If yes, please describe briefly the national benefits.

The Centre of Australian Category Theory at Macquarie University is an internationally recognized mathematical research hub which attracts respect for Australian achievements. This leads to worldwide confidence in Australia's status as important culturally, socially and economically.

Part E RESEARCH COLLABORATION

E1

Provide details of any collaborations or partnerships the research involved or led to – for each, as appropriate, list the names of the partner organisation, its type and country of location.

Name of Inst/Org	Type of Inst/Org	Country (if not Australia)
Department of Pure Mathematics and Mathematical Statistics, University of Cambridge	University	UK
Department of Mathematics and Statistics, Masaryk University	University	Czech Rep.
Department of Mathematics, University of Chicago	University	USA
Department of Mathematics, Université Montpellier 2	University	France
École de Mathématique, Université catholique de Louvain	University	Belgium
Institute for Particle and Nuclear Physics, Budapest	Research institute	Hungary
Department of Mathematics and Statistics, Dalhousie University	University	Canada

Part F REPORT ON COLLABORATION BY INDUSTRY PARTNER/s

To be completed and certified by each Industry Partner.

You may attach a more extended statement if necessary.

Part F is not applicable to DP scheme.

Part G PROJECT OUTPUTS

G1 Field of research

<i>RFCD code</i>	<i>% weighting</i>
230104	100

G2 Socio-economic objective

<i>SEO code</i>	<i>% weighting</i>
780101	100

G3 Publications and other academic outputs

Enter the number of outputs in each category. For programs other than LIEF, enter full details; for publications, include 'published' and 'in press' publications, but exclude 'forthcoming' and 'submitted' work.

Number

A1 Book—authored research

2

1. R.H. Street, Quantum Groups: a path to current algebra, Australian Math. Society Lecture Series 19 (Cambridge University Press; 18 January 2007; ISBN-978-0-521-69524-4).
2. Dominic Verity, Complicial sets: characterising the simplicial nerves of strict omega-categories. *Memoirs Amer. Math. Soc.* 193 (2008), no. 905, xvi+184 pp.

A2 Book—authored other

A3 Book—edited

5

1. Davydov, M. Batanin, M. Johnson, S. Lack and A. Neeman “Categories in algebra, geometry and mathematical physics.” Papers from the conference and workshop in honor of Ross Street's 60th birthday held at Macquarie University, Sydney, July 11-16, 2005 and the Australian National University, Canberra, July 18-21, 2005. *Contemporary Mathematics* 431 (American Mathematical Society, Providence, RI, 2007; xii+467 pp. ISBN: 978-0-8218-3970-6; 0-8218-3970-5).
2. Michael Johnson and Varmo Vene (eds). *Algebraic Methodology and Software Technology. Lecture Notes in Computer Science* 4019 (Springer-Verlag, 2006), 387pp.
3. Jawad Abuhlail, Gabrielle Böhm, Tomasz Brzezinski, H. Peter Gumm, Salah-Eddine Kabbaj, Ross Street, Robert Wisbauer (editors), “Interactions of Algebraic and Coalgebraic Structures (Theory and Applications)” *The Arabian Journal for Science and Engineering* 33(2C) (December 2008) 585pp.
4. Ross Street, Maria Manuel Clementino and Samson Abramsky (editors) Special Issues arising from CT2007 at Carvoeiro (Algarve, Portugal): “Topology and Homotopy” *Applied Categorical Structures* 17(3) (June 2009); “Semi-abelian Categories” *Applied Categorical Structures* 17(4) (August 2009); “Toposes and Functor Categories” *Applied Categorical Structures* 18(2) (April 2010); “Enriched and Higher Categories” *Applied Categorical Structures* 18(3) (June 2010).
5. Martin Hyland, George Janelidze, Michael Johnson, Peter Johnstone, Stephen Lack, Ross Street, Walter Tholen and Richard Wood) Special Issue: Dedicated to the Memory of Professor Gregory Maxwell Kelly, *Applied Categorical Structures* 19(1) (February 2011).

A4 Book—revision/new edition

A4 Book—translation

B Book chapter

2

1. Michael Johnson, Enterprise Software with Half-Duplex Interoperations, In Doumeingts, Mueller, Morel and Vallespir (eds), *Enterprise Interoperability: New Challenges and Approaches* (Springer-Verlag, 2007) 521–530.
2. Michael Johnson, Mathematical support for ensemble engineering, *LNCS* 5380 (Springer 2008) 179–189.

C1 Journal article—articles in scholarly refereed journal

1. D.R. Verity, Weak Complicial Sets II, Nerves of Complicial Gray Categories, in: "Categories in Algebra, Geometry and Mathematical Physics", Contemporary Mathematics 431 (American Math. Soc. 2007) 441-467.
2. B.J. Day and E. Panchadcharam and R.H. Street, Lax braidings and the lax centre, in "Hopf Algebras and Generalizations" Contemporary Mathematics 441 (2007) 1-17.
3. B.J. Day and R.H. Street, Centres of monoidal categories of functors, in "Categories in Algebra, Geometry and Mathematical Physics", Contemporary Mathematics 431 (2007) 187-202.
4. E. Panchadcharam and R.H. Street, Mackey functors on compact closed categories, Journal of Homotopy and Related Structures 2(2) (2007) 261–293;
<http://arxiv.org/abs/0706.2922>.
5. E. Panchadcharam and R.H. Street, Pullback and finite coproduct preserving functors between categories of permutation representations: corrigendum and addendum, Theory and Applications of Categories 18(5) (2007) 151-156.
6. Ross Street, "Gregory Maxwell Kelly 1930-2007", Historical Records of Australian Science 21(2) (CSIRO Publishing, 2010) 237-251.
7. M. Johnson and R. Rosebrugh, Fibrations and Universal View Updatibility, Theoretical Computer Science 388 (2007) 109-129.
8. S.G. Lack, Bicat is not triequivalent to Gray, Theory Appl. Categ. 18 (2007) 1-3.
9. B.J. Day and S.G. Lack, Limits of small functors, J. Pure Appl. Algebra 210 (2007) 651-663.
10. R. Cockett and S.G. Lack, Restriction categories. III. Colimits, partial limits and extensivity, Math. Structures Comput. Sci. 17 (2007) 775-817.
11. Simona Paoli, Semistrict models of connected 3-types and Tamsamani's weak 3-groupoids. J. Pure Appl. Algebra 211 (2007) 801-820.
12. Thorsten Palm, Categories with Slicing, Theory and Applications of Categories 22 (2009) 97–135.
13. Glynn Winskel, Event structures with symmetry; in: "Computation, meaning, and logic: articles dedicated to Gordon Plotkin", Electron. Notes Theor. Comput. Sci. 172 (Elsevier, Amsterdam, 2007) 611-652.
14. Craig Pastro and Ross Street, Doubles for monoidal categories, Theory and Applications of Categories, 21(4) (2008) 61–75.
15. Brian Day and Craig Pastro, Note on Frobenius monoidal functors, New York J. Math. 14 (2008) 733–742.
16. Dominic Verity, A tricategorical approach to change of base in category theory, Reprints in Theory and Application of Categories, 256 pp. (accepted 3 March 2008)
17. Stephen Lack and Simona Paoli, 2-nerves for bicategories. K-Theory 38 (2008) 153–175.
18. Peter Johnstone, Stephen Lack and Pawel Sobocinski, Quasitoposes, quasiadhesive categories and Artin glueing. Algebra and coalgebra in computer science, Lecture Notes in Comput. Sci., 4624 (Springer, Berlin, 2007) 312–326.
19. Stephen Lack, Homotopy-theoretic aspects of 2-monads. J. Homotopy Relat. Struct. 2 (2007) 229–260.
20. Dominic Verity, Weak complicial sets. I. Basic homotopy theory, Advances in Math. 219 (2008) 1081–1149.
21. Ross Street, An Australian conspectus of higher categories, in: Towards Higher Categories, edited by John C. Baez and J. Peter May, The IMA Volumes in Mathematics and its Applications, Volume 152 (Springer, 2009) 237-264.
22. Craig Pastro and Ross Street, Weak Hopf monoids in braided monoidal categories,

- Algebra and Number Theory 3(2) (2009) 149-207.
23. Craig Pastro and Ross Street, Closed categories, star-autonomy, and monoidal comonads, *Journal of Algebra* 321(11) (Special Issue in Honor of Gus Lehrer, 2009) 3494-3520.
 24. Ignacio Lopez-Franco, Ross Street and Richard Wood, Duals invert, *Applied Categorical Structures* 19(1) (special volume in memory of Max Kelly, February 2011) 321-362.
 25. Ross Street, Weak distributive laws, *Theory and Applications of Categories* 22(12) (2009) 313-320.
 26. Dimitri Chikhladze, Barr's Embedding theorem for enriched categories, *Journal of Pure and Applied Algebra* 215 (2011) 2148-2153.
 27. Dimitri Chikhladze, Category of quantum categories, *Theory and Applications of Categories* 25 (2011) 1-37.
 28. Dimitri Chikhladze, Stephen Lack and Ross Street, Hopf monoidal comonads, *Theory and Applications of Categories* 24 (2010) 554-563.
 29. Dimitri Chikhladze, The Tannaka representation theorem for separable Frobenius functors, *Algebras and Representation Theory* (accepted on 22 February 2011).
 30. David Oury, On the duality between discs and trees, *Theory and Applications of Categories* 16 (2010) 418-450.
 31. Stephen Lack and John Power, Gabriel-Ulmer duality and Lawvere theories enriched over a general base, *Journal of Functional Programming* 19 (2009) 265-286.
 32. Stephen Lack, A 2-categories companion, in: *Towards Higher Categories*, edited by John C. Baez and J. Peter May, *The IMA Volumes in Mathematics and its Applications*, Volume 152 (Springer, 2009) 105-192.
 33. Stephen Lack and Jiri Rosicky, Notions of Lawvere theory, *Applied Categorical Structures*, 19 (2011) 363-391.
 34. Michael Johnson and Robert Rosebrugh, Ontology engineering, universal algebra and category theory, in *Theory and Applications of Ontologies volume 2: The Information-science Stance* (edited by Roberto Poli, Achilles Kameas and Michael Healy, Springer-Verlag, to appear).
 35. Michael Johnson, John Power and David Naumann, Category Theoretic Models of Data Refinement, *Electronic Notes in Theoretical Computer Science* (Elsevier Science, 2009) 21-38.
 36. Michael Johnson and Stefano Kasangian, A Relational Model of Incomplete Data without NULLs, *Journal of Research and Practice in Information Technology* (to appear).
 37. Ross Street and Dominic Verity, The comprehensive factorization and torsors, *Theory and Applications of Categories* 23 (2010) 42-75.
 38. Micah Blake McCurdy and Ross Street, What separable Frobenius monoidal functors preserve, *Cahiers de topologie et géométrie différentielle catégoriques* 51 (2010) 29-50.
 39. Stephen Lack, Note on the construction of free monoids, *Applied Categorical Structures* 18 (2010) 17-29.
 40. Richard Garner, Homomorphisms of higher categories, *Advances in Math.* 224 (2010) 2269-2311.
 41. Stephen Lack, Icons, *Applied Categorical Structures* 18 (2010) 289-307.
 42. Stephen Lack, R.F.C. Walters, and R.J. Wood, Bicategories of spans as cartesian bicategories, *Theory and Applications of Categories* 24 (2010) 1-24.
 43. Stephen Lack, A Quillen model structure for Gray-categories, *Journal of K-theory* (accepted 11 Jun 2010).
 44. Alain Bruguières, Stephen Lack and Alexis Virelizier, Hopf monads on monoidal categories, *Advances in Mathematics* (accepted 23 Nov 2010).
 45. Gabriella Böhm, Stephen Lack, and Ross Street, Weak bimonads and weak Hopf

- monads, *Journal of Algebra* 328 (2011) 1–30.
46. Thomas Booker and Ross Street, Torsors, herds and flocks, *Journal of Algebra* 330 (2011) 346–374 (also see <http://arxiv.org/abs/0912.4551>).
47. Carolyn Kennett, Emily Riehl, Michael Roy, Michael Zaks, Levels in the toposes of simplicial sets and cubical sets, *Journal of Pure and Applied Algebra* (published online December 2010).
48. Brian J. Day, Monoidal functor categories and graphical Fourier transforms, *Theory and Applications of Categories* 25 (2011) 118–141.
49. Stephen Lack and Jiri Rosicky, Enriched weakness, *Journal of Pure and Applied Algebra* (accepted 24 Jan 2011).
50. Emily Riehl, On the structure of simplicial categories associated to quasi-categories, *Mathematical Proceedings of the Cambridge Philosophical Society* (accepted 6 December 2010).
51. Stephen Lack and Jiri Rosicky, Notions of Lawvere theory, *Applied Categorical Structures* (accepted 14 Oct 2009).
52. Richard Garner, Ionads, *Journal of Pure and Applied Algebra* (to appear).
53. Benno van den Berg and Richard Garner, Types are weak omega-groupoids, *Journal of the London Mathematical Society* (to appear).
54. Benno van den Berg and Richard Garner, Topological and simplicial models of identity types, *Transactions of the ACM on Computational Logic* (to appear).

C2 Journal article—other contribution to refereed journal

C3 Journal article—non-refereed article

1

W. Chen, K. Matthews and R. Street, Dr Peter Pleasants, *Gazette Austral. Math. Society* 35 #3 (July 2008) 162–165.

C4 Journal articles—letter or note

D Major reviews

1

Ross Street, Review of: Eckmann-Tierney eds: *Seminar on Triples and Categorical Homology Theory* (Repr. *Theory Appl. Categ.* No. 18 (2008) 303 pp.); in *Mathematical Reviews* MR2410935.

E1 Conference—full written paper-refereed proceedings

2

1. Michael Johnson and Robert Rosebrugh, Constant Complements, Reversibility and Universal View Updates, *Lecture Notes in Computer Science* 5140 (Springer 2008) 238–252.
2. Michael Johnson and Robert Rosebrugh, Implementing a categorical information system, *Lecture Notes in Computer Science* 5140, (Springer 2008) 232–237.

E2 Conference—full written paper-non-refereed proceedings

E3 Conference—extract of paper

E4 Conference—edited volume of conference proceedings

E4 Conference—unpublished presentation

21

1. Stephen Lack, Strict morphisms and weak morphisms, invited plenary lecture at CT08 in Calais, France (June 2008).
2. Ross Street, Quantum categories, invited plenary lecture at CT08 in Calais, France (June 2008).
3. Stephen Lack, Notions of Lawvere theory, invited talk at the Max Kelly Memorial Conference, Cape Town (Jan 2008).
4. Michael Johnson, Max Kelly and Category Theory, invited opening talk at the Max Kelly Memorial Conference, Cape Town (Jan 2008).
5. Ross Street, Categorical aspects of Mackey functors, invited talk at the Mathematics Institute of the Autonomous University of Barcelona (July 2008).
6. Ross Street, "What separable Frobenius monoidal functors preserve", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
7. Stephen Lack, "Welcome to weak world", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
8. Dominic Verity, "Weak complicit sets and iterated Segal spaces", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
9. Dominic Verity, "The comprehensive factorisation and torsors" (Category Theory Seminar, University of Cambridge, UK; 13 October 2009).
10. Dominic Verity, "On the canonicity of complicit stratifications" (PSSL88 conference in honour of Martin Hyland and Peter Johnstone, University of Cambridge, UK; 3-5 April 2009).
11. John Bourke, "Exact 2-categories", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
12. Micah Blake McCurdy, "Graphical methods for the Tannaka constructions", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
13. Dimitri Chikhladze, "Quantum categories", CT2009 International Conference on Category Theory (Cape Town, 29 June - 4 July 2009).
14. Ross Street, "From linear algebra to knot theory" (Colloquium, University of Adelaide, August 2009).
15. Ross Street, "Weak Hopf algebras and Frobenius algebras" (Differential Geometry Seminar, University of Adelaide, August 2009).
16. Tom Booker and David Oury, "Category Theory", displayed at the Macquarie University Faculty of Science Research Conference (2 October 2009).
17. Steve Lack, "A Quillen model structure for Gray-categories", Some Trends in Algebra '09 (Prague 7-11 Sept 2009)
18. Steve Lack, "A Quillen model structure for Gray-categories", (Topology Seminar, Ecole Polytechnique Federale de Lausanne, 18 Sept 2009)
19. Steve Lack, "My life as a category theorist", Annual Meeting of the Australian Mathematical Society (Adelaide, 28 Sept - 1 Oct 2009)
20. Richard Garner, "Lax aspects of abstract homotopy theory", Category Theory 2010 (Genova, Italy, July 2010).
21. Stephen Lack, "Hopf monads", Category Theory 2010 (Genova, Italy, July 2010).

F Audio-visual recording

G Computer software

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H Designs

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J1 Major creative works

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J2 Creative work included in group exhibition, performance, recording or anthology

--

J3 Exhibition curatorship

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K Other academic outputs (in categories other than those listed above)

25

1. Elango Panchadcharam, Categories of Mackey Functors (PhD Thesis, Macquarie University, approved 10 April 2007).
2. Steve Bennoun, Tannaka Duality for Comodules over a Hopf Algebra (Masters Thesis at the École Polytechnique Fédérale de Lausanne, March 2007; supervised by R. Street).
3. Thomas Booker, Non-Standard Numbers as Global Sections of a Dedekind Reals Object (Honours Essay at Macquarie University, December 2007; supervised by R. Street).
4. Dimitri Chikhladze, "Elements of enriched and quantum category theory" (Macquarie University, submitted 2 July 2010; PhD awarded 28 September 2010; supervisor Street).
5. John Bourke, "Codescent objects in 2-dimensional universal algebra" (University of Sydney, submitted 30 August 2010; PhD awarded February 2011; supervisor Lack).
6. B.J. Day, Association schemes and classical RCFTs as "graphic" Fourier transformations. $\langle \text{math.CT/0702208} \rangle$ 4 pages CT(QA).
7. B.J. Day, Biclosed bicategories: Localisation of convolution. $\langle \text{math.CT/07053485} \rangle$ 4 pages CT.
8. B.J. Day and Craig Pastro, On endomorphism algebras of separable monoidal functors. $\langle \text{math.CT/07121864} \rangle$ 17 pages CT(QA).
9. R.H. Street, Obituary of Professor G. Maxwell Kelly, "Polymath revelled in the mystery of numbers" Sydney Morning Herald (Timelines, 11 April 2007).
10. R.H. Street, Obituary of Professor G. Maxwell Kelly, Gazette Austral. Math. Society 34 #2 (May 2007) 94-96.
11. Brian J. Day, A star-autonomous category of Banach spaces—Correction (arXiv:0901.4403).
12. Brian J. Day, Note on the fusion map (arXiv:0902.2259).
13. Brian J. Day, Note on analytic functors as Fourier transforms (arXiv:0906.3330).
14. Brian J. Day, When is existential quantification conservative? (arXiv:0906.4594).
15. Brian J. Day and Craig A. Pastro, Note on endomorphism algebras of separable monoidal functors (arXiv:0907.3259).
16. Brian J. Day, On the existence of category bicompletions (arXiv:0909.5010).
17. John Bourke, The category of categories with pullbacks is cartesian closed (arXiv:0904.2486).
18. Dominic Verity, Change of Base in Enriched and Internal Category Theory", invited for Reprint Theory and Applications of Categories.
19. Dominic Verity, Weak complicial sets III - Iterated Segal Spaces and Coherence.
20. Dominic Verity, Weak complicial sets IV - Simplicial Cobordisms.
21. Stephen Lack, Clubs, wreaths, and ordinals.

22. Tom Booker and Alexei Davydov, Commutative algebras in Fibonacci categories.
23. Dominic Verity, "Relating Descent Notions" open letter to Urs Schreiber, http://golem.ph.utexas.edu/category/2009/07/verity_on_descent_for_strict_g.html
24. Stephen Lack, Non-canonical isomorphisms (submitted; also arXiv:0920.2996).
25. Richard Garner, An abstract view on syntax with sharing (submitted to Oxford Journal of Logic and Computation).

G4 Have any of the investigators or their works won prizes, awards or other major tributes as a result of the research project? ☒ Yes
If yes, give details.

Stephen Lack was awarded the Medal of the Australian Mathematical Society for 2009.

Ross Street was awarded the 2011 Chaire de la Vallée Poussin by the Catholic University of Louvain (Louvain-la-neuve, Belgium).

G5 Research commercialisation

Provide details of commercialisation resulting from the research project against the categories listed below.

Commercialisation Types

Patent patents establish legally enforceable protection of rights over intellectual property associated with inventions)				
Patents filed				
Type	Where lodged	Application number	Title	
Patents pending				
Type	Where lodged	Application number	Title	
Invention disclosure an invention disclosure occurs when a device, substance, method or process that is apparently new, useful and involves an inventive step is made known to personnel within an institution who have responsibility for managing the institution 's patenting and research commercialisation activities				
				Invention disclosures
Plant breeder right a plant-breeder right is a temporary monopoly granted to a plant breeder seeking registration of a new plant variety. It gives the breeder the exclusive right to produce and reproduce the propagating material of the variety and to stock, sell, import or export the propagating material. In order for a plant-breeder right to be granted, the variety must be novel, distinct, uniform and stable.				
				Plant Breeder rights
Application number for each filed		Certificate number for each granted		
Licence executed (include licences, options and assignments) A licence agreement formalises the transfer of technology between two parties, where the owner of the technology (the licensor) permits the other party (the licensee)to share the rights to use the technology. An option agreement grants the potential licensee a time period during which it may evaluate the technology and negotiate the terms of a licence agreement. An assignment agreement conveys all right, title and interest in and to the licensed subject matter to the named assignee.				
				Licences executed
Start-up company formed <i>'start-up companies' refers to companies engaged in businesses that were dependent, for their formation, upon licensing or assignment of technology developed in this research project.</i>				
				Start-up companies formed
Name of company			ABN	
Describe any other commercial outcomes from the project				

- G6 Evidence of impact and contribution. Is there evidence that this project has had an impact in the research field or the broader public domain? ☒ Yes
If yes, briefly describe the impact and contribution.

The ARC has access to standard citation data on articles published in ISI journals. However, there may be other indicators of impact including, for example, citations to books, re-publication, translations, reviews, invited keynote addresses, other invitations, newspaper/media /expert commentary or advice to Government.

Dominic Verity's PhD thesis was accepted for re-publication.
All CIs were on numerous conference scientific committees.
All CIs were invited keynote speakers at various conferences (see Item G3 part E4).

Part H RESEARCH TRAINING, CAREERS AND EMPLOYMENT

H1 Postgraduate research training – Australian postgraduates supported by this project

Stipends: include students receiving ARC-funded stipends under Discovery and Linkage, but exclude DEST Australian Postgraduate Awards (APAs)

Research support: include students whose postgraduate or Honours research is assisted by ARC funding (for example, use of equipment funded under LIEF, international research experience funded under Linkage-International, and project support provided under Discovery-Projects or Linkage-Projects)

Number of PhD students receiving stipends and project support	0
Number of research Masters students receiving stipends and project support	0
Number of PhD students receiving project support but not stipends	3
Number of Masters students receiving project support but not stipends	0
Number of Honours students receiving project support but not stipends	2

H2 Postgraduate research training – Overseas postgraduates supported by this project

Research support: include students enrolled in overseas universities whose postgraduate or Honours research is assisted by ARC funding (for example, use of equipment funded under LIEF, international research experience funded under Linkage-International)

Number of overseas PhD students involved in the project	8
Number of overseas research Masters students involved in the project	2
Number of overseas Honours students involved in the project	0

H3 Early career researchers

Early career researchers are researchers with up to 5 years postdoctoral experience

ECRs named in the application
no ECRs named in the application

Did any other ECRs participate? If so, how many? Not selected

H4 Other employed personnel

Number of research associates/assistants funded (full time)	1
Number of research associates/assistants funded (part time)	1
Number of professional and/or technical officers	
Industry Partner employees (not PIs)	
Other personnel involved (provide details below)	21
Visiting Researchers: Glynn Winskel (Jan 2007; Cambridge University) Clemens Berger (Feb 2007; Laboratoire J.A. Dieudonné, Université de Nice-Sophia Antipolis, France) Robert Rosebrugh (Aug 2007; Mt Allison U, Canada) Dorette Pronk (Jul 2007; Dalhousie University, Halifax, Canada) Jiri Rosicky (Nov 2007; Masaryk University, Czech Republic)	

Alexis Virelizier (14 Nov-14 Dec 2008; University of Montpellier II)
Alain Bruguières (14 Nov-14 Dec 2008; University of Montpellier II)
Martin Markl (18 Nov-15 Dec 2008; Institute of Mathematics of the Academy of Sciences of the Czech Republic)
Denis-Charles Cisinski (30 Aug-27 Sep 2008; University of Paris XIII)
Mark Weber (30 Jul-7 Sep 2008; University of Paris VII)
Ezra Getzler (July 2008; Northwestern University)
Michael Shulman (May 2008; University of Chicago)
Peter Johnstone (Apr 2008; Cambridge University)
Boris Chorny (Mar 2008; Australian National University)
Isar Stubbe (Feb-Mar 2008; University of Antwerp)
Richard Wood (Jan-Feb 2008; Dalhousie University, Halifax, Canada)
Clemens Berger (Feb 2009; Laboratoire J.A. Dieudonné, Université de Nice-Sophia Antipolis, France)
Robert Rosebrugh (Feb-Mar 2009; Mt Allison U, Canada)
Stefano Kasangian (Dipartimento di Matematica "Federigo Enriques", Università degli Studi di Milano, Italy)
Richard Garner (Oct 2009-Feb 2010; University of Cambridge)
Emily Riehl (Oct 2009-Feb 2010; University of Chicago)
Gabriella Böhm (Jan-Feb 2010; Research Institute for Particle and Nuclear Physics, Budapest)
Marcelo Aguiar (Feb 2010; Texas A&M)

Part I FUNDING CONTRIBUTIONS

Part I is not applicable to DP scheme.

Part J OTHER FUNDING

J1 Income

[Include only those income sources identified in the application form]

Source - DIRECT COSTS only	Request/commitment in application	Actual contribution
ARC Discovery funding	462827	287199
University/ies		
Other (provide details in section below relating to Actual 'Other' contribution only)		
Salaries for CIs	459858	450000
TOTAL - DIRECT COSTS only	922685	737199

J2 Has funding from other sources, including other ARC grants, supported this research?

Yes

If yes, provide details of the source and amount of funding received, under the following headings:

	Details of source/grant type	Funding received (\$AUS)
Host Institution		
	Total Host Institution support	0
ARC Funding Programs		
	Total other ARC Funding Programs	0
Other Government Competitive Funds		
	Total other Government competitive funds	0
Other Government Non-Competitive Funds		
	Total other Government Non-Competitive funds	0
Other funds from industry		
	Total other industry funds	0
Other external support		
	Morgan-Phoa Donation	30000
	Total other external support	30000
	GRAND TOTAL	30000

Part K

Thank you for completing the final report.

The ARC may contact you if clarification or further information is required to acquit the grant.

Information on this form is collected in order to determine whether the research project funded by the ARC has reached satisfactory completion and for post award reporting.

Researchers should note that if the Final Report is not submitted on time, or if the ARC is not satisfied with the outcomes of the Project, this will be noted against any further Applications under any ARC program submitted by, or on behalf of, the Chief Investigator or Fellow and will be taken into account in the assessment of those applications. Applications under any ARC program will be deemed ineligible if submitted by, or on behalf of a Chief Investigator or Fellow on a Project for which the Final Report is outstanding.

Information in the Final Reports enables the ARC to account for the public funding of research and to promote the value of research to the Australian community. The ARC compiles statistical reports on the outcomes of publicly-funded research, drawing on the information you have provided in this Final Report.

Drawing on the information in the Final Report on the results, outcomes and significance of the research, the ARC may identify projects which have broad interest for media coverage and/or inclusion in ARC publications including the ARC website. The ARC will endeavour to contact you should it wish to publicise the results of your research, but bear in mind that this may not always be possible.